

MATH GR5450: Credit Analytics

Instructors: David X. Li and Steven Zhu

Time: Spring 2026, Friday 1:10 pm-3:40 pm

Location: 401 Chandler

Office Hours: by appointment

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Course Overview

This course addresses an important subject area in Financial Mathematics that is covered only sporadically in existing courses in the MAFN program. It will give deep and comprehensive coverage of the technical foundations of credit analytics as applied to credit risk, credit derivatives, and counterparty credit risk management. This involves survival analysis, stochastic hazard rate modelling, copula functions, and loss distributions (typically used in insurance modelling), as well as applications of machine learning to alternative data, for example, in analyzing exposure from credit cards and loans. The course will also cover the models and estimation methodologies designed to measure and manage counterparty credit risk for individual products and portfolios. The course will give students/graduates an advantage in jobs where they deal with credit risk, credit derivatives, counterparty risk modelling, or applications of machine learning to credit analysis.

Prerequisites

Prerequisites: Required: Math GR5010 Intro to the Math of Finance (or equivalent). Recommended: Stat GR5264 Stochastic Processes – Applications I (or equivalent).

Course Materials

1. Reference Books

Modelling Single-Name and Multiple-Name Credit Derivatives, Dominic O’Kane, Wiley 2008, ISBN 9080470519288.

Credit Risk: Pricing, Measurement, and Management, Darrell Duffie and Kenneth J. Singleton, Princeton University Press, 2003, ISBN 9780691090467.

Counterparty Credit Risk and Credit Value Adjustment, John Gregory, Wiley Finance Series 2012, John Wiley & Sons Ltd. ISBN 978-1118316641

2. Lecture Notes: will be provided in class.

3. Reading Materials: will be distributed before the class. A partial list is here:

- a. CreditMetrics Technical Document, The RiskMetrics Group, 1997 (CTD 1997)
- b. Constructing a credit Curve, David X. Li, 1998, Risk (DL 1998)
- c. On default correlation: a copula function approach, David X. Li, 2000 (DL2000)

- d. Single Tranche CDOs, David X. Li and Jure Skarabot, 2003 (DL2003)
- e. Valuation of Single Name CDS with risky Counterparty, 2015(DL2015)
- f. An Overview on Copula Function Methods in Credit Portfolio Modelling, David X. Li, 2008 (DL2008)
- g. A Guide to Modelling Counterparty Credit Risk, Michael Pykhtin, Steven Zhu, 2007 (PZ 2007)
- h. ISDA CDS Standard Model (2021) (ISDA 2021)
- i. "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates". Journal of Finance. 29 (2): 449–470. Merton, Robert C. (1974). (RB 1974)
- j. Stress Testing Transition Matrix, Joonwoo Kim (2000) (JK 2000)
- k. Finding Generators for Markov Chains via Empirical Transition Matrix, Jason Z. Wei, 2000 (JW 2000)

Grading

Your grades will be determined according to the following formula:

<i>Item</i>	<i>Description</i>	<i>Date due</i>	<i>Method of submission</i>	<i>Percentage</i>
<i>Attendance</i>	<i>Attend class unless excused in advance</i>	<i>Every class</i>	<i>In-Person</i>	<i>20%</i>
<i>Homework</i>	<i>Assignments (4)</i>		<i>written</i>	<i>40%</i>
<i>Two Exams</i>	<i>The first exam will cover the first half, and the second one cover the second half</i>		<i>written</i>	<i>40%</i>
<i>Total</i>				<i>100%</i>

Class Participation

Students are expected to read the materials before attending classes and actively participate in class discussions.

Homework

There will be 4 problem sets. Students can discuss these problems, but they must submit their own assignment. Direct copying of each other's homework is prohibited.

Academic Integrity

Students must exhibit the highest level of personal and academic honesty and integrity in the course. You must always submit your work and not that of others, except where group work is explicitly allowed or encouraged. When in doubt, consult the professor. Infractions will have serious consequences under the university's rules.

Disabilities accommodations

If you have been certified by Disability Services to receive accommodation, please ask your liaison in GSAS to consult with your professor.

Course Outline

1. Empirical studies of historical default probability, recovery rate, and rating migration matrix

Reading Materials:

- DS Chapter 3, 4
- Moody's and S&P historical default studies

2. Basics of survival analysis: survival function, hazard rates

Reading Materials

- DL (1998)
- DS Chapter 3

3. Valuation of risky bonds: one-period model (Duffie Singleton), and multiple-period model, and Valuation of Single-name Credit: CDS

Reading Materials

- DL (1998)
- DS Chapter 5
- DL (2012)
- ISDA (2021)

4. Default correlation: discrete approach and survival time correlation, basics of Copula models

Reading Materials

- DL (2012)
- ISDA (2021)
- DL (2000)

5. Credit Portfolio models and Implementation

Reading Materials

- DL (2003, 2008)
- CTD (1997)

6. CDO valuation, base correlation curve

Reading Materials

- DL (2003, 2008)

7. Exam 1

8. Counterparty Credit Risk (CCR) – Overview and Learns Learned

9. CCR – Modeling Credit Exposure at Transaction Level for Derivative Products

10. CCR – Modeling Credit Exposure at Portfolio Level: Netting and Collateral

11. CCR – Wrong Way Risk and Counterparty Default Loss Simulation

12. CCR – Credit Valuation Adjustments and XVA
13. Central Clearing Counterparty (CCP): Margin and Collateral Modeling
14. Regulatory Framework for Counterparty Credit Risk

Class	Selected Topics
Class 1 – 1/23	Empirical studies of historical default probability, recovery rate, and rating migration matrix
Class 2 – 1/30	Basics of survival analysis: survival function, hazard rates
Class 3 – 2/06	Valuation of risky bonds: one-period model (Duffie Singleton), multiple-period model, and Valuation of Single-name Credit: CDS
Class 4 – 2/13	Default correlation: discrete approach and survival time correlation, basics of Copula models
Class 5 – 2/20	Credit Portfolio Models and Implementation
Class 6 – 2/27	CDO valuation, base correlation curve
Class 7 – 3/06	Midterm Exam
Class 8 – 3/13	Counterparty Credit Risk (CCR) – Overview and Learnings Learned
Class 9 – 3/20	Spring Break
No Class – 3/27	CCR – Modelling Credit Exposure at Transaction Level for Derivative Products
No Class – 4/03	CCR – Modelling Credit Exposure at Portfolio Level: Netting and Collateral
Class 10 – 4/10	CCR – Wrong Way Risk and Counterparty Default Loss Simulation
Class 11 – 4/17	CCR – Credit Valuation Adjustments and XVA
Class 12 – 4/24	Central Clearing Counterparty (CCP): Margin and Collateral Modelling
Class 13 – 5/01	Regulatory Framework for Counterparty Credit Risk
Class 14 – 5/08	Final Exam - Group Project Presentation